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Large vessel occlusion identification network with vessel guidance and asymmetry learning on CT angiography of acute ischemic stroke patients

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ABSTRACT

Identifying large vessel occlusion (LVO) is of significant importance for the treatment and prognosis of acute ischemic stroke (AIS) patients. CT Angiography (CTA) is commonly used in LVO identification due to its visibility of vessels and short acquisition time. It is challenging to make LVO identification methods focus on vascular regions without vessel segmentation while accurate vessel segmentation is difficult and takes more time. Meanwhile, most existing methods fail to effectively integrate clinical prior knowledge. In this work, we propose VANet, a novel LVO identification network which utilizes coarse-grained vessel feature for feature enhancement and learns asymmetry of two brain hemispheres on CTA of AIS patients. Firstly, we reconstruct 3D CTA scans into 2D based on maximum intensity projection (MIP) to reduce computational complexity and highlight vessel information. Secondly, we design a coarse-grained vessel aware module based on simple edge detection and morphological operations to acquire coarse-grained vessel feature without precise vessel segmentation. Thirdly, we design a vessel-guided feature enhancement that directs the model's attention to vessel areas in the images by utilizing coarse-grained vessel feature. Finally, inspired by the clinical knowledge that LVO can lead to asymmetry in brain, we design an asymmetry learning module utilizing deep asymmetry supervision to keep the patients' inherent asymmetry invariant and using asymmetry computing to acquire effective asymmetry features. We validate the proposed VANet on our private internal and external AIS-LVO datasets which contain 366 and 81 AIS patients, respectively. The results indicate that our proposed VANet achieves an accuracy of 94.54% and an AUC of 0.9685 on the internal dataset, outperforms 11 state-of-the-art methods (including general classification methods and LVO-specific methods). Besides, our method also achieves the best accuracy of 88.89% and AUC of 0.9111 when compared to 11 methods on the external test dataset, implying its good generalization ability. Interpretability analysis shows that the proposed VANet can effectively focus on vascular regions and learn asymmetry features.

1. Introduction

Stroke is the second leading cause of death in the world and the leading cause of death in China (Johnson et al., 2016; Wu et al., 2019). Acute Ischemic Stroke (AIS) accounts for approximately 70% of all incident stroke cases in China, as documented by the National Epidemiological Survey of Stroke in China (NESS-China) (Wu et al., 2019). Large vessel occlusion (LVO) is a sub-type of AIS with higher disability and mortality rates (Malhotra et al., 2017; Lakomkin et al., 2019). Furthermore, the treatment and prognosis of LVO are highly time-dependent. Timely and accurate LVO identification, a binary classification task (LVO vs. non-LVO), can help doctors make better treatment which is of great significance to improve survival rate and prognosis of LVO patients.

Computed Tomography Angiography (CTA) is a non-invasive medical imaging technique. CTA data is typically acquired after the injection of contrast agents to the patients. These contrast agents enhance the visibility of blood vessels, making them easier to differentiate and aiding in the detection of vascular diseases or abnormalities. And the time required for AIS patients to undergo a head CTA examination is relatively short. Therefore, in clinical practice CTA is commonly used for LVO identification (Mayer et al., 2020; McTaggart et al., 2017).

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Deep learning methods show promising performance in the automated diagnosis of LVO (Shlobin et al., 2022; Ghozy et al., 2023). LVO identification requires attention to vascular regions in CTA. Vessel segmentation can help the model focus on vessels, thereby enhancing the performance of LVO identification. Therefore, vessel segmentation is widely applied to existing automated LVO identification methods (Stib et al., 2020; Su et al., 2020; Yahav-Dovrat et al., 2021). However, vessel segmentation in 3D CTA is difficult due to the anatomical complexity, wide intensity distribution and small volume proportion of vessels (Sun et al., 2023). And vessel segmentation in 3D CTA is computational expensive and time-consuming. Inaccurate vessel segmentation may reduce the performance of LVO identification. Hence, it is necessary to design an LVO identification method that does not rely on accurate vessel segmentation. Considering the contrast of vessels in CTA, it is feasible to extract coarse-grained vessel feature when there are no accurate vessel segmentation labels. Therefore, in order to reduce the negative impact of vessel segmentation on LVO identification, we aim to design a method which can achieve coarse-grained vessel feature extraction and direct model's attention to vessel regions without accurate vessel segmentation.

Clinical prior knowledge suggests that bilateral symmetry information can be utilized to the LVO identification. Morphologically speaking, a normal human head exhibits a high level of bilateral symmetry, although it is not perfectly symmetrical (Maes et al., 1999). The degree of asymmetry has long been thought to be helpful for suggesting a pathological condition and/or providing a diagnostic cue for clinicians (Liu, 2009). For example, abnormal asymmetry in the brain indicates a wide range of pathologies, such as stroke, bleeding and tumor. In the task of LVO identification using CTA, abnormalities may arise in the affected brain region due to interrupted or reduced blood flow supply caused by LVO. Significant differences in symmetry between left and right brain region may suggest the possibility of LVO. DeepSymNet analyzed the symmetry difference between two hemispheres in 3D CTA images for LVO identification (Barman et al., 2019). However, focusing exclusively on differences in two hemispheres of brain may lead the model to overlook the features of vessel in the image. In deep learning model, symmetrical images may lose their symmetry after feature extraction because all feature maps are globally pooled before inputting them into the classification layer. Inconsistency in symmetry before and after feature extraction makes it difficult for model to learn and utilize effective symmetry features. Hence, to address these issues, it is necessary to learn effective asymmetry features to improve the performance of LVO identification.

In this paper, we propose an automated LVO identification method named Vessel-guided and Asymmetry learning Network (VANet) that directs attention to the vascular regions without requiring accurate vessel segmentation and can acquire bilateral asymmetry features as suggested by clinical prior knowledge. In preprocessing, we project original CTA scans into 2D images based on maximum intensity projection (MIP) algorithm to reduce computational cost and create a clearer view of vessels. Initially, we propose a Coarse-grained Vessel Aware Module (CVAM) to get coarse-grained vessel feature. Then, we design Vessel-Guided Feature Enhancement (VGFE) based on attention mechanism which utilizes coarse-grained vessel feature to guide the model's attention towards vascular regions. Subsequently, we design an Asymmetry Learning Module (ALM), utilizing our designed deep asymmetry supervision to ensure that feature symmetry aligns with clinical prior knowledge after feature extraction and computing asymmetry features for fusion with original features to enhance the model's representation learning capabilities. We validate our proposed VANet on both our private internal and external AIS-LVO datasets. Experimental results demonstrate that VANet outperforms 11 state-of-the-art methods across multiple evaluation metrics on both internal and external AIS LVO datasets.

The contributions of our work are summarized as follows:

- We propose a novel deep learning network named VANet for LVO identification that utilizes coarse-grained vessel feature to guide the model's attention, extracts effective asymmetry features to enhance the performance of LVO identification inspired by clinical prior knowledge.
- We design a Coarse-grained Vessel Aware Module based on edge detection and morphological operations, extracting coarse-grained spatial location information of vessels.
- We design a Vessel-Guided Feature Enhancement module based on attention mechanism which utilizes captured coarse-grained vessel feature to enhance the learning ability of vessel-related features.
- We design an Asymmetry Learning Module that can preserve inherent asymmetry of patients' data after feature extraction and compute asymmetry features.

2. Related work

In this section, deep learning methods available for LVO identification and general image classification will be reviewed.

2.1. Automated LVO identification methods

Considering that LVO is a vascular disorder, vessel segmentation is widely applied in the preprocessing of LVO diagnosis (Yahav-Dovrat et al., 2021; Stib et al., 2020; Su et al., 2020). A recent study explored the automatic identification of LVO using multi-phase CTA with preprocessing based on 3D vessel segmentation and Max Intensity Projection (MIP), and achieved better model performance on multi-phase CTA than single-phase (Stib et al., 2020). The algorithm for LVO identification in some medical products, such as Viz LVO and Rapid LVO, also involves vessel segmentation (Yahav-Dovrat et al., 2021; Delora et al., 2024). Vessel segmentation can enhance the signal-to-noise ratio of input data by removing non-vascular tissue. However, accurate vessel segmentation is a more difficult task than LVO identification. And the results of vessel segmentation will greatly affect the accuracy of LVO identification. DeepSymNet analyzed the difference between two hemispheres in 3D CTA images to deduce if the patient has suffered from AIS without needing the location of the areas affected (Barman et al., 2019). However, focusing exclusively on the differences in two hemispheres of brain may make it more difficult for the model to learn effective vessel features. At the same time, detecting LVO in 3D CTA is computationally expensive.

2.2. General image classification methods

In recent years, deep learning has made significant strides in natural image classification tasks, with many methods based on CNN (Chen et al., 2023; Ding et al., 2022; Wang et al., 2024; Ding et al., 2024; Woo et al., 2023) or Vision Transformer (ViT) (Dosovitskiy et al., 2020; Liu et al., 2021; Zhu et al., 2023; Li et al., 2023) demonstrating outstanding performance. After reconstructing 3D CTA into 2D images based on MIP, these natural image classification methods can be applied to LVO identification. Some studies aimed to develop universally applicable medical image diagnostic methods (Isensee et al., 2021; Cheng et al., 2023; Huang et al., 2023). MedVit aimed to enhance the model's robustness and generalization capabilities for application in medical image classification (Manzari et al., 2023). In automated medical image diagnosis, models that incorporate relevant clinical prior knowledge often experience considerable performance improvements (Barman et al., 2019; Kuang et al., 2021). Lin et al. (2023) re-grouped the input images based on the structural correlation between different image modalities and achieved excellent performance. However, these mentioned methods directly applied to LVO identification are unable to effectively integrate task-specific prior knowledge, leading to limitation of the performance of certain specific tasks. Therefore, the integration of prior knowledge for task-specific methods still have good application prospects.

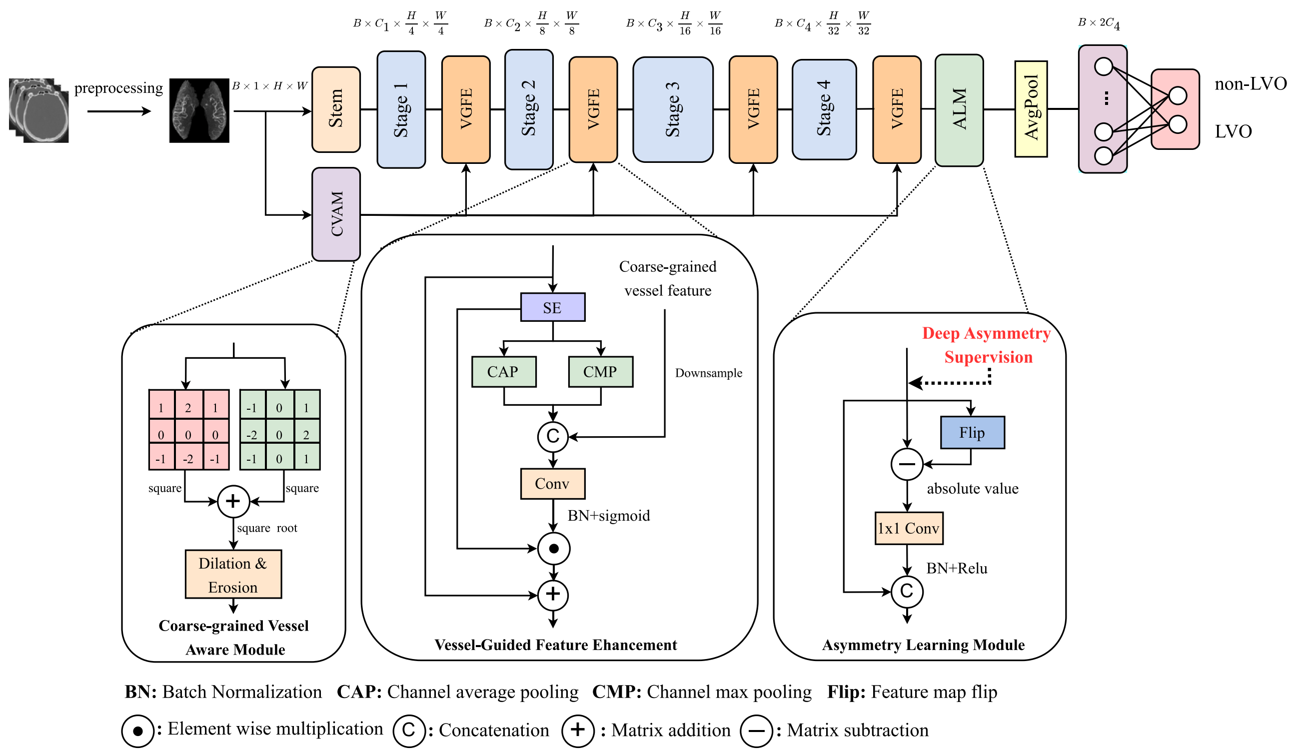

Fig. 1. Overview of our proposed method. It includes preprocessing, Stem and four stages from backbone and our designed coarse-grained vessel aware module, vessel-guided feature enhancement after each stage and Asymmetry Learning Module. Preprocessing steps include skull-stripping, registering CTA scans to the Montreal Neurological Institute (MNI) space, Middle Cerebral Artery (MCA) region generation and Maximum Intensity Projection (MIP). The Stem and 4 stages are the same as the backbone: RepVit. The three subplots show the details of our designed modules. The coarse-grained vessel aware module contains sobel convolution for edge detection, dilation and erosion. The vessel-guided feature enhancement consists of channel attention and spatial attention guided by coarse-grained vessel feature. The Asymmetry Learning Module is used for acquiring and utilizing effective asymmetry features. Global average pooling and fully connected layer are used as classifier to get the final classification results.

3. Methodology

In this section, we will introduce the overall architecture and technical details of our proposed VANet.

3.1. Overview

The pipeline of our proposed VANet is shown in Fig. 1. Firstly, we perform a series of preprocessing operations on the initial 3D CTA data to obtain a 2D reconstructed image, and then input the image into the proposed model. We utilize RepVit (Wang et al., 2024) as the backbone of our method, a standard lightweight CNN based on depthwise separable convolution and squeeze-and-excitation module (SE) (Hu et al., 2018), incorporating the architectural designs of lightweight ViTs. The Stem module and stages in Fig. 1 are the same as RepVit (Wang et al., 2024). We firstly design CVAM based on edge detection and morphological operations to acquire coarse-grained vessel feature and then perform fusion in VGFE after each stage to direct the model's attention towards vessel area. Finally, we design the ALM which utilizes our designed deep asymmetry supervision to keep the semantic asymmetry observed in patients align with clinical prior knowledge thereby extracting effective asymmetry features.

3.2. Data preprocessing

In data preprocessing, each original CTA undergo a series of preprocessing steps before it is used as input to our model. Firstly, we follow a skull-stripping operation based on convex optimization with shape propagation proposed by Najm et al. (2019) and utilize its published code[1] to extract brain tissues from acquired CTA images in a slice-by-slice fashion. Subsequently, the images are registered to the Montreal Neurological Institute (MNI) space (Evans et al., 2012) using affine transformation to reduce the impact of asymmetry caused by inherent factors and data acquisition angle. Filtering the vascular regions of interest can reduce the effects of overlapping large vessels and the presence of visible venous phases caused by delayed acquisition (Kumar et al., 2023). Therefore, we register the standard vessel territory template proposed by Liu et al. (2023) to the obtained MNI images of all patients using affine transformation for generating the individual vessel territory maps where we then can extract the Middle Cerebral Artery (MCA) region of all patients. Then, in order to create a clearer visualization of the vessel regions in CTA and reduce computational complexity, we select 25% slices whose center slice is the one contain the largest brain region and then these selected slices are processed by MIP in the axial direction to obtain MIP images with a size of 256×256. Then, we perform z-score normalization on the MIP images.

3.3. Coarse-grained vessel aware module

Accurate vessel segmentation in 3D CTA is difficult and computational expensive. We assume that coarse-grained vascular region localization can also assist doctors in diagnosis and guide the deep learning model to automatically learn vascular features for LVO identification. Since vessels have obvious edge features in CTA, the features extracted by edge detection operators are spatially correlated to vessels and can help locate the approximate vessel regions. In the input layer, we use edge detection operation based on sobel convolution kernel to

[1] https://github.com/WuChanada/StripSkullCT.

acquire edge feature. Firstly, we acquire the horizontal and vertical edge feature by using the horizontal Sobel kernel and the vertical Sobel kernel. Followed by computing square root of sum of squares of horizontal and vertical edge, we obtain the final spatially vessel-related edge feature. And the whole process can be described as:

$$X_H = K_H(X), X_V = K_V(X),$$
$$Edge(X) = \sqrt{(X_H)^2 + (X_V)^2}, \tag{1}$$

where K_H, K_V denotes the horizontal and vertical Sobel kernels. X_H and X_V denotes the horizontal and vertical edge information. $Edge(X)$ denotes the edge feature of the input.

Subsequently, inspired by vessel segmentation methods that utilize edge detection (Lesage et al., 2009), we perform a series of operations aimed at aligning the feature distribution with that of vessels. Specifically, we apply morphological operations including dilation and erosion to the edge features obtained, thereby achieving gap filling. The dilation and erosion can be computed as:

$$(f \oplus K)(x, y) = \max_{(s,t)\in K} \{f(x+s, y+t)\}, \tag{2}$$

$$(f \ominus K)(x, y) = \min_{(s,t)\in K} \{f(x+s, y+t)\}, \tag{3}$$

where f denotes the input feature, K denotes the dilation or erosion kernel, $\oplus$ denotes the dilation operation, $\ominus$ denotes the erosion operation, (x, y) denotes the coordinates of an element in the feature and (s, t) denotes the relative coordinates within the range of the kernel. The dilation operation is to take the maximum value from elements of the input feature covered by kernel K as the value of the corresponding element of the output feature, while the erosion operation takes the minimum value. And the combination of dilation and erosion can be computed as:

$$f \bullet K = (f \oplus K) \ominus K, \tag{4}$$

where $\bullet$ denotes the morphological closing operation which can achieve gap-filling to make the feature more spatially vessel-related.

3.4. Vessel-guided feature enhancement

Attention mechanisms are often used to guide models to focus on important features in channels or spaces. Therefore, we use the coarse-grained feature and combine it with attention mechanisms to guide the model's attention towards vascular regions. Our designed VGFE is a residual architecture with a hybrid attention branch. In the hybrid attention branch, we firstly apply the Squeeze-and-Excitation (SE) module (Hu et al., 2018) for channel attention to learn the importance of different channels in the features. Subsequently, we fuse the spatially vessel-related feature for fusion to generate spatial attention. In the spatial attention part, we firstly concatenate features from channel average pooling (CAP) and channel max pooling (CMP) to get the spatial mask at different scales from current semantic layer. And then we downsample the vessel-related feature from edge-aware module to the same size and perform a concatenation operation. Followed by the convolution layer, batch normalization and sigmoid activation function, the spatial attention mask is generated. And we use broadcast element wise multiplication to adjust the model's spatial attention. The whole process of VGFE can be described as

$$Attn = Sigmoid(Conv([CAP(SE(X)), CMP(SE(X)), V])), \tag{5}$$

where CAP and CMP denotes the channel average pooling and channel max pooling, V denotes coarse-grained vessel feature after downsampling and $Sigmoid$ denotes sigmoid activation function.

3.5. Asymmetry learning module

Clinical prior knowledge suggests that CTA images with LVO demonstrate semantic asymmetry while those of healthy brains exhibit semantic symmetry. Hence, in order to learn effective asymmetry features, we design the ALM which contains deep asymmetry supervision and asymmetry computing module. The deep asymmetry supervision is to keep the patients' inherent asymmetry invariant after feature extraction. And asymmetry computing module aims to effectively utilize asymmetry features.

3.5.1. Deep asymmetry supervision

Obtaining symmetry features requires ensuring the effectiveness of the features. However, LVO identification task of deep learning often does not pay attention to the geometric properties of features. In terms of task characteristics, the desired final result only includes the class to which the object belongs. From a model implementation perspective, the deepest feature maps are globally pooled and then passed to the classification layer. In this process, all geometric properties include symmetry are erased. And we consider symmetry or asymmetry to be a special global feature. It is difficult for model to acquire this semantic geometric property. Therefore, we propose deep asymmetry supervision to acquire effective asymmetry features.

Firstly, to supervise features at the semantic level based on symmetry, we introduce a metric named Asymmetry Degree (AD) to quantitatively describe the symmetry of feature maps. The calculation principle is to calculate the distance between the feature map and its self-flip feature map. The calculation of AD for feature maps can be expressed as

$$AD = \frac{1}{c \times h \times w} \sum_{i=1}^{c} \sum_{j=1}^{h} \sum_{k=1}^{w} (y_{ijk} - \hat{y}_{ijk})^2, \tag{6}$$

where $\hat{y}$ denotes the feature map flipped along the dimension of w, c, h and w denotes the channel, height and width of features. A lower AD value suggests a higher degree of symmetry, whereas a higher AD value suggests a higher degree of asymmetry. Considering the distinction between LVO and non-LVO. We assign different computation methods to the loss function for each sample based on the labels. The purpose of this approach is to preserve semantic symmetry in images of non-LVO individuals while reinforcing the asymmetry in images of LVO. This aligns with the trends observed in AD according to clinical prior knowledge. After obtaining the AD to quantitatively describe the semantic symmetry, our loss function for deep asymmetry supervision is designed as follows:

$$f(AD) = \begin{cases} e^{-AD}, & \text{if } label = 0 \\ 1 - e^{-AD}, & \text{if } label = 1, \end{cases} \tag{7}$$

$$L_{sym} = \frac{1}{b} \sum_{i=1}^{b} f(AD_i), \tag{8}$$

where $label = 0$ denotes the samples without LVO which means semantic symmetry, $label = 1$ means semantic asymmetry for the presence of LVO and b represent the batch size of features. Since the deep symmetry supervision plays an auxiliary role in our task, we constrain the function in the calculation method to vary between 0 and 1. In conclusion, for the features in $B \times C \times H \times W$ format, our proposed deep first calculates the AD value of each sample along dimension of B, and then give samples different calculation methods based on actual labels. More specifically, the function f exhibits an increasing trend when applied to the AD of LVO samples, whereas it shows a decreasing trend for the AD of non-LVO samples. So we can achieve the purpose of ensuring the semantic symmetry of non-LVO while enhancing the semantic asymmetry of LVO.

Table 1
Demographic information for the internal dataset and external dataset.

Characteristic	Internal dataset (n = 366)	External dataset (n = 81)
Sex, female (%)	172 (47.0%)	35 (43.2%)
Age, median (IQR)	72 (62–82)	73 (62–80)
LVO (%)	221 (60.4%)	60 (74.1%)
Time from onset to CTA (h), median (IQR)	2.1 (1.4–2.8)	1.5 (1.0–3.4)
NIHSS, median (IQR)	14 (7–19)	13 (6–16)
ASPECTS, median (IQR)	10 (8–10)	8 (6–10)

3.5.2. Asymmetry computing module

We consider symmetry to be a special global feature that cannot be easily acquired by normal feature extractor. Therefore, we specially design the ALM to obtain symmetry features and fuse symmetry features with original features. The axis of symmetry is important for obtaining symmetry features. In the preprocessing step, we perform linear registration on the original CTA data. Although there are symmetry differences caused by factors not related to LVO which have a negative impact on our use of asymmetry information for LVO identification, we believe that computing symmetry in the semantic layer can reduce this negative impact. So, in the semantic level, we can use the self-flip subtraction to acquire symmetry feature. The process of calculating symmetry feature can be expressed as

$$F = Conv([X, Conv(|X - Flip(X)|)]), \tag{9}$$

where $|X - Flip(X)|$ denotes the symmetry feature that calculated by self-flip subtraction, $[.]$ denotes concatenation, and $Conv$ represents a 1×1 convolution layer followed by batch normalization and ReLU activation. In the ALM module, we simply fuse the symmetry features and the original features through a concatenation operation, and then globally pool the fused features and input them into the classification layer.

3.6. Total loss function

During the model training process, the LVO identification loss function uses class-weighted binary cross-entropy loss, and the computation of classification loss and total loss is detailed as follows:

$$L_{cls} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_{y_i} \cdot y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)\right], \tag{10}$$

$$L_{total} = \alpha \cdot L_{sym} + \beta \cdot L_{cls}, \tag{11}$$

where L_{cls} denotes cross entropy loss with class weights, α and β denotes the weights of different loss function. In our task, the classification loss is dominant and the deep asymmetry supervision plays an auxiliary role. So we constrain the range of the loss function value in the calculation method of the deep asymmetry supervision. And then we simply set the weight of each loss function to 0.5.

4. Experiments and results

4.1. Dataset collection

Private internal AIS-LVO dataset: Our private internal dataset includes single-phase CTA data from 366 AIS patients, among which 221 are identified as LVO cases, obtained from Xiangya Hospital, Central South University. Each patient's diagnosis is jointly provided by several experienced radiologists to ensure the accuracy. We provide detailed clinical information of the dataset we used, which includes patient age, gender, acquisition time from symptom onset to CTA image, National Institutes of Health Stroke Scale (NIHSS), Alberta Stroke Program Early CT Score (ASPECTS). Details are shown in Table 1. We conduct experiments on this dataset for internal validation, using a 5-fold cross validation for the spilt.

Private external AIS-LVO dataset: Our external private dataset includes single-phase CTA data from 81 AIS patients, with 60 cases identified as LVO. This dataset is sourced from the First Affiliated Hospital, University of South China. The detailed demographic information is shown in Table 1. In the experiments, we utilize this dataset for external validation.

4.2. Implementation details

All experiments are implemented by PyTorch with two NVIDIA RTX 3090 GPUs on Ubuntu 18.04.5 LTS. We conduct LVO identification experiments on internal datasets, using a random 5-fold cross-validation for the split of internal dataset. We select 3, 1 and 1 folds as train, validation and test sets, and repeat cross-validation experiment 5 times to obtain LVO identification results of all patients. We select the model with the best performance in cross-validation for external validation. The size of each input data in all datasets used is 256 × 256 with a single channel. In our codes, the data preprocessing section utilizes linear registration based on Antspy, data reading and writing, and MIP operations use SimpleITK. We incorporate random Gaussian noise, random elastic transformation and random flipping provided by the kits from MONAI transform for data augmentation in the training set. For more details, please refer to our codes at Github.[2]

4.3. Comparison methods

The methods for comparison include 8 methods for general natural image classification: FasterNet (Chen et al., 2023), RepLKNet (Ding et al., 2022), ConvNextV2 (Woo et al., 2023), UniRepLKNet (Ding et al., 2024) , UniFormer (Li et al., 2023), RVT (Mao et al., 2022), BiFormer (Zhu et al., 2023) and RepVit (Wang et al., 2024), and 1 general medical image classification method: MedVit (Manzari et al., 2023), and 2 special methods for LVO identification: (Stib et al., 2020) and DeepSymNet (Barman et al., 2019). For fair comparison, we reproduce all compared methods on the same train, validation and test sets respectively in each 5-fold cross-validation experiment to obtain their results. Specifically, for (Stib et al., 2020) and DeepSymNet (Barman et al., 2019) without official codes, we carefully follow the technical details including preprocessing steps and model design described in the published papers to reproduce them. For the remaining 9 methods with official codes, we directly use the official codes for reproduction. Note that the hyperparameter settings for all compared methods are optimized on the same validation sets in each 5-fold cross-validation experiment.

4.4. Evaluation metrics

LVO identification is a binary classification problem. We use Accuracy (ACC), Area Under the Curve (AUC), Precision, Recall and F1 score to quantitatively evaluate the classification performance. AUC is the area under the Receiver Operating Characteristic (ROC) curve, which measures the classification ability of the model. Precision is used to measure the proportion of true positives among the samples predicted as positive by the model. Recall focuses on the model's ability to capture true positives among the actual positive instances. F1 score maintain a balance between Precision and Recall and is suitable for imbalanced classes. The calculation of evaluation metrics can be described as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \tag{12}$$

$$Precision = \frac{TP}{TP + FP}, \tag{13}$$

[2] https://github.com/lxyuanyyyy/VANet.

Table 2
Quantitative results of VANet and 11 methods on our internal AIS-LVO dataset. The best and second metrics are in bold and underlined respectively.

Model	ACC (%)	AUC	Precision	Recall	F1	Params(M)	FLOPs(G)
FasterNet (Chen et al., 2023)	85.52 ± 3.73	0.8891 ± 0.0363	0.8564 ± 0.0620	0.9142 ± 0.0541	0.8839 ± 0.0316	29.90	5.95
RepLKNet (Ding et al., 2022)	92.08 ± 0.77	0.9614 ± 0.0089	0.9241 ± 0.0105	0.9456 ± 0.0116	0.9348 ± 0.0089	78.84	20.33
ConvNextV2 (Woo et al., 2023)	89.62 ± 1.02	0.9257 ± 0.0059	0.9181 ± 0.0237	0.9097 ± 0.0089	0.9134 ± 0.0084	27.80	5.81
UniRepLKNet (Ding et al., 2024)	91.26 ± 1.68	0.9574 ± 0.0144	0.9460 ± 0.0206	0.9026 ± 0.0280	0.9236 ± 0.0219	55.56	12.10
UniFormer (Li et al., 2023)	89.89 ± 1.68	0.9320 ± 0.0261	0.9179 ± 0.0249	0.9147 ± 0.0172	0.9159 ± 0.0188	21.03	4.49
RVT (Mao et al., 2022)	91.26 ± 2.04	0.9574 ± 0.0124	0.9100 ± 0.0422	**0.9500 ± 0.0181**	0.9287 ± 0.0175	21.76	5.61
BiFormer (Zhu et al., 2023)	89.34 ± 1.02	0.9282 ± 0.0059	0.9095 ± 0.0237	0.9146 ± 0.0089	0.9118 ± 0.0084	25.02	5.48
RepVit (Wang et al., 2024)	89.07 ± 2.04	0.9230 ± 0.0248	0.9099 ± 0.0080	0.9103 ± 0.0319	0.9097 ± 0.0156	**4.72**	**1.10**
MedVit (Manzari et al., 2023)	90.44 ± 4.09	0.9522 ± 0.0220	0.9019 ± 0.0530	0.9495 ± 0.0156	0.9242 ± 0.0293	31.14	7.72
Stib et al. (2020)[a]	87.43 ± 0.77	0.9194 ± 0.0169	0.8768 ± 0.0231	0.9237 ± 0.0161	0.8992 ± 0.0073	6.95	3.68
DeepSymNet (Barman et al., 2019)[a]	91.53 ± 1.02	0.9373 ± 0.0148	0.9397 ± 0.0255	0.9184 ± 0.0085	0.9286 ± 0.0092	5.09	1.59
Ours(VANet)	**94.54 ± 0.39**	**0.9685 ± 0.0047**	**0.9640 ± 0.0210**	0.9453 ± 0.0201	**0.9541 ± 0.0048**	4.96	1.11

[a] Denotes the methods specifically designed for LVO identification.

$$Recall = \frac{TP}{TP + FN}, \tag{14}$$

$$F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}, \tag{15}$$

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\theta)\, d\mathrm{FPR}(\theta), \tag{16}$$

where TP denotes True Positive, TN denotes True Negative, FP denotes False Positive and FN denotes False Negative. In Eq. (16), TPR denotes true positive rate, FPR denotes false positive rate and θ is the classification threshold. AUC calculates the area between TPR and FPR when θ varies from 0 to 1.

4.5. Comparison with other methods

4.5.1. Results on the internal dataset

We employ a diverse set of methods to compare with our proposed VANet, including specialized approaches for LVO identification, models designed for medical image classification tasks and models for natural image classification. The experimental results are shown in Table 2. The results demonstrate that our proposed VANet achieves superior performance across multiple metrics with the second smallest parameter sum. From Table 2, we can find that our proposed VANet achieves an ACC of 94.54%, an AUC of 0.9685 and an F1 of 0.9541 with a parameter of 4.96M. And it is worth noting that the numerical difference between the Recall of our method and the best performance of Recall is extremely small. The FLOPs of VANet is 1.11G which is almost equally the lowest as RepVit while the ACC is over 5% higher than RepVit. Among all the classification methods for natural image compared, RepLKNet (Ding et al., 2022) achieves the second best accuracy, AUC and F1 while with a parameters of 78.84M which is much higher than our method. The performance of Stib et al. (2020) is not satisfactory, and we believe that this is due to the negative impact of poor vessel segmentation results in its preprocessing. DeepSymNet (Barman et al., 2019) also achieves excellent performance with a relatively low number of parameters, surpassing most natural image classification methods with much higher number of parameters which proves the effectiveness of methods designed specifically for LVO identification. And our proposed method achieves better performance with fewer parameters than DeepSymNet. In conclusion, our proposed VANet achieves a well-balanced trade-off between performance and parameters.

4.5.2. Results on the external test dataset

To verify the generalization ability of our proposed method, we conduct experiments on our private external dataset. From Table 3 we can find that our proposed method achieves the best ACC (88.89%), AUC (0.9111) and F1 (0.9256), outperforming all compared methods. DeepSymNet and our proposed VANet outperform other methods on multiple important metrics, demonstrating excellent generalization ability of the methods specially designed for LVO identification. Although our VANet does not achieve optimal performance in Recall and Precision. However, when considering the metric Recall and Precision, it becomes apparent that the F1 score for both the RepVit (Wang et al., 2024) method with the best Recall and Stib et al. (2020) with the best Precision is much lower than our method. This suggests that there is an inherent bias in these two methods, which results in inflated metrics for either Precision or Recall. In contrast, our method achieves a good balance between the Precision and Recall. Comprehensive results show that our proposed method achieve the best performance, proving its good generalization ability.

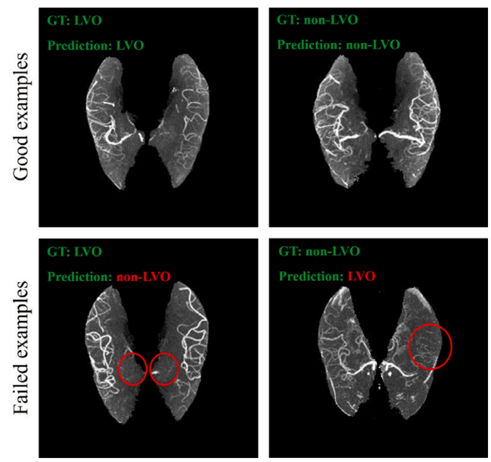

Fig. 2. Visualization of VANet's successful and failed samples. GT denotes ground truth. The first row shows two correctly identified samples. The second row shows two misclassified samples.

4.6. Visualization analysis of VANet

We have visualized some success and failure samples of our proposed method, as shown in Fig. 2. For the correctly identified LVO sample, visible large vessels can be seen on the left side but not on the right side, and the difference between left and right is obvious which indicates the presence of LVO. For the correctly classified non-LVO sample, the vessels on both sides are bright and clear, demonstrating there is no LVO on both sides. For the misclassified LVO sample, although some large vessels disappear in both sides, other vessels on both sides are relatively clear and the bilateral difference is small which causes our model to incorrectly classify it as non-LVO. As for the misclassified non-LVO sample, although some large vessels can be seen in both sides, many small vessels disappear or become dark in the left side, causing asymmetry between the left and right hemispheres,

Table 3
Performance of different model in external test dataset. The best and second metrics are in bold and underlined respectively.

Model	ACC (%)	AUC	Precision	Recall	F1
FasterNet (Chen et al., 2023)	75.31	0.7222	0.8448	0.8167	0.8305
RepLKNet (Ding et al., 2022)	85.19	0.8833	0.9286	0.8667	0.8965
ConvNextV2 (Woo et al., 2023)	80.24	0.8778	0.8928	0.8333	0.8621
UniRepLKNet (Ding et al., 2024)	83.95	0.8619	0.8730	0.9167	0.8943
UniFormer (Li et al., 2023)	81.48	0.8635	0.8689	0.8833	0.8760
RVT (Mao et al., 2022)	83.95	0.9087	0.8507	0.9500	0.8976
MedVit (Manzari et al., 2023)	82.72	0.8714	0.8382	0.9500	0.8906
BiFormer (Zhu et al., 2023)	76.54	0.8587	<u>0.9362</u>	0.7333	0.8224
RepVit (Wang et al., 2024)	82.72	0.8198	0.8194	**0.9833**	0.8939
(Stib et al., 2020)[a]	74.07	0.8492	**0.9535**	0.6833	0.7961
DeepSymNet (Barman et al., 2019)[a]	<u>85.19</u>	<u>0.8841</u>	0.8529	<u>0.9667</u>	<u>0.9063</u>
Ours	**88.89**	**0.9111**	0.9180	0.9333	**0.9256**

[a] Denotes the methods specifically designed for LVO identification.

and leading to misclassification of our model. These results imply that both considering the occlusion status of large vessels and small vessels may further improve the performance of LVO identification.

4.7. Ablation study

We conduct ablation study to examine the role of each component of VANet on our private internal AIS-LVO dataset.

4.7.1. Effectiveness of CVAM

In order to explore the effectiveness of our proposed CVAM, we conduct two experiments. First, we remove the CVAM from proposed VANet and only use the channel pooling operations to generate the spatial attention mask without performing fusion with coarse-grained vessel feature. The spatial attention without the guidance of vessel feature can refer to the spatial attention part of convolutional block attention module (Woo et al., 2018). Second, we only add our proposed CVAM to the backbone: RepVit where we apply the sigmoid activation function to the output of CVAM for attention. As shown in Table 4, the experiments indicate a noticeable decrease in the model's ACC without the CVAM and coarse-grained vessel feature. The ACC drops by 1.10% while AUC drops by 0.082. And after only adding the proposed CVAM to the backbone, the model's ACC and AUC increase by 1.64% and 0.0262 compared to the backbone. The results demonstrate the effectiveness of coarse-grained vessel feature extracted by our proposed CVAM.

4.7.2. Effectiveness of VGFE

To prove the effectiveness of our proposed VGFE, we test the performance of proposed model without our proposed VGFE module or only adding VGFE to the backbone. After removing VGFE from VANet, we apply the sigmoid activation function to the outputs of CVAM and multiply them with original features, then input the results to the following stages. As for only adding VGFE to the backbone, we generate the spatial attention mask only using the outputs of channel pooling without concatenating coarse-grained vessel features. As shown in Table 4, the performance of the model drops, with ACC and AUC dropping by 1.42% and 0.0197 respectively. And the ACC and AUC of backbone integrating VGFE increase by 1.91% and 0.0204. These results show that our use of coarse-grained vessel feature to guide the model's attention is effective.

4.7.3. Effectiveness of ALM

In order to explore the effectiveness of ALM, we perform experiments that remove the ALM from our model or only add the ALM in the backbone. The results are shown in Table 4. The results indicate that the model without ALM achieves accuracy of 92.90% which is 1.64% lower than the model with ALM. And the performance of the backbone also improves after adding the ALM. The results prove the effectiveness of our proposed ALM.

Table 4
Ablation study of our proposed method on the internal dataset. The first row adding none of three modules denotes the backbone: RepVit.

Module			ACC (%)	AUC	Params(M)
CVAM	VGFE	ALM			
			89.07	0.9230	4.72
✓			90.71	0.9492	4.72
	✓		90.98	0.9434	4.81
		✓	91.26	0.9400	4.86
	✓	✓	93.44	0.9603	4.96
✓		✓	93.12	0.9488	4.86
✓	✓		92.62	0.9457	4.81
✓	✓	✓	**94.54**	**0.9685**	4.96

5. Discussion

In this paper, we propose a method for LVO identification based on clinical prior knowledge. In order to direct the model's attention towards vessels without vessel segmentation, we introduce vessel-guided feature enhancement by utilizing coarse-grained vessel feature. To extract and fuse effective symmetry features, we design an asymmetry learning module which involves deep asymmetry supervision and asymmetry computing module to acquire semantic bilateral difference features. Additional discussions are listed.

5.1. Interpretability analysis

5.1.1. Interpretability analysis of VANet and comparison methods

To analyze the interpretability of our proposed VANet, we use the Grad-CAM (Selvaraju et al., 2017) algorithm to visualize the attention heatmap of our proposed VANet and compare it with other baseline methods. The visualization samples are shown in Fig. 3. For fairness, we select layers with consistent relative positions of these models as the target layer of Grad-CAM algorithm. As indicated by the red arrows in the second row, VANet is able to focus on the vessels in the right hemisphere where LVO occurs, while other methods may not pay sufficient attention. And in the first row, our VANet demonstrates heightened attention towards areas of vessel distribution compared to other models, particularly focusing on thicker and brighter vessels, aligning with the clinical prior knowledge of LVO and the characteristics of CTA data. In summary, the visualization results demonstrate that our VANet model effectively attends to the blood vessel distribution area in CTA, enhancing the interpretability of our LVO identification task.

5.1.2. Interpretability analysis of CVAM

We visualize the heatmap of our proposed CVAM, as shown in Fig. 4. The visualization results demonstrate that CVAM can locate the vessel location well, and it pays more attention to large and brighter vessels and relatively less attention to small and darker vessels is lower. This

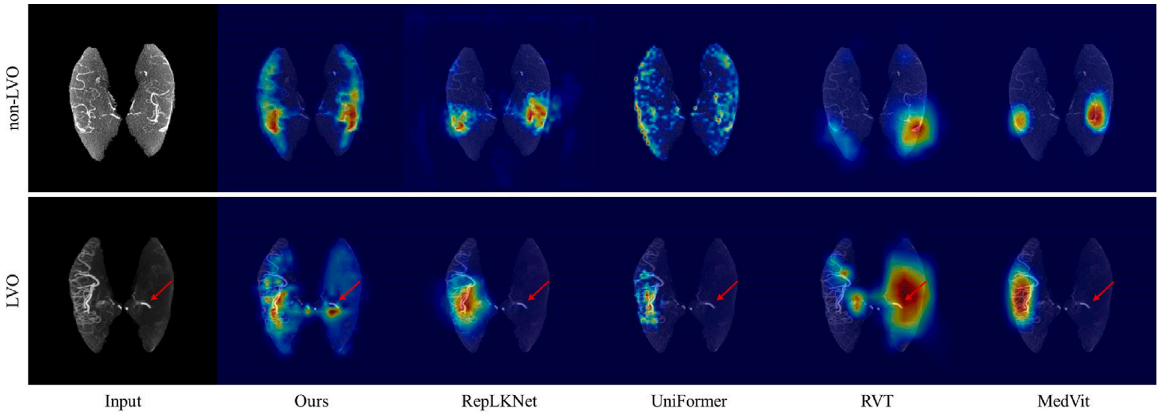

Fig. 3. Visualization samples of heatmaps from our proposed method and other baseline methods. The red arrow points to some bright vessels. Fewer vessels are visible in the hemisphere with LVO.

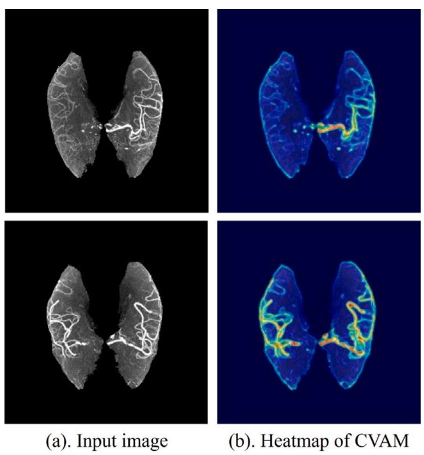

Fig. 4. The heatmaps of coarse-grained vessel aware module. (a) shows the preprocessed MIP image. (b) shows the heatmaps of coarse-grained vessel aware module.

indicates that CVAM is capable of effectively extracting coarse-grained vascular features. Therefore, guiding the model with the output of CVAM to focus on vascular regions has good interpretability.

5.1.3. Interpretability analysis of VGFE

In this work, we design VGFE to direct the model's attention towards vessel regions. To analyze the interpretability of our designed VGFE, we utilize the Grad-CAM (Selvaraju et al., 2017) algorithm to visualize the heatmaps of models with and without VGFE. From the visualization samples shown in Fig. 5 we can find that the model with VGFE can better focus on vascular regions while pays more attention to brighter vessels. Model without VGFE does not pay insufficient attention to the vessels on the both side. Combined with visualization examples, our designed VGFE achieves the purpose of guiding the model to focus on the vascular regions which also proves that VGFE can improve the performance of the model as the experimental results indicate.

5.1.4. Interpretability analysis of ALM

Inspired by clinical knowledge, we design ALM to learn semantic symmetry features of LVO and non-LVO. In order to analyze the interpretability of our designed ALM, we visualize the heatmaps of models with and without ALM. As shown in Fig. 6, the model tends to focus on the side with more and brighter vessels for LVO samples. For non-LVO sample in row 2, models with ALM can better focus on information from both sides of the brain while the model without ALM focus more on the left hemisphere, potentially resulting in incorrect identification. The visualization samples illustrate that our designed ALM aligns with clinical prior knowledge and improves the performance and interpretability of the model.

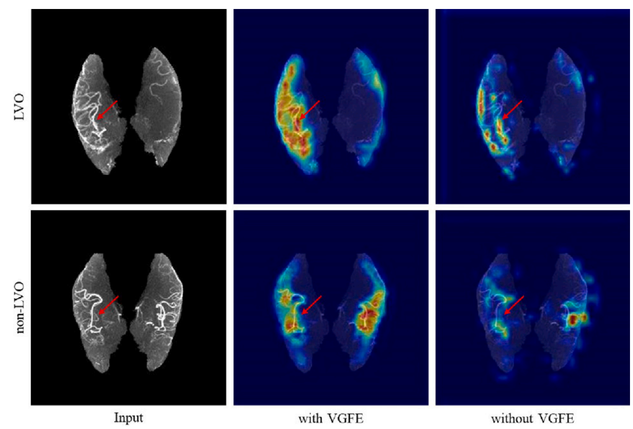

Fig. 5. Visualization samples of model with or without VGFE. The red arrow points to some bright vessels.

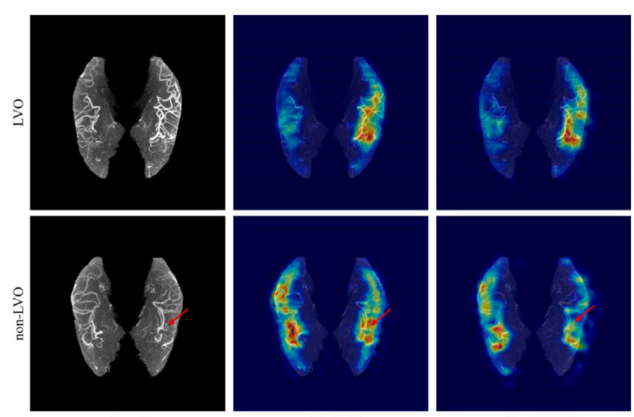

Fig. 6. Visualization samples of model with or without ALM. The red arrow points to some bright vessels.

5.1.5. Clinical evaluation of our vanet

To validate the proposed method's applicability in real-world clinical settings we invite 3 radiologists with 5 or more years of experience

Table 5
The quantitative results of clinical feedback from 3 radiologists. ACC denotes accuracy.

Radiologist	ACC of using CTA (%)	ACC of using heatmap (%)	ACC of using both CTA and heatmap (%)	ACC of vessel attention (%)	ACC of asymmetry consistency (%)
A	94.26	94.26	94.81	96.44	98.09
B	93.4	93.99	93.99	95.36	96.72
C	94.54	94.26	95.01	96.17	98.63

Table 6
The comparison with 3D classification methods.

Model	ACC (%)	AUC
DeepSymNet(3D version)	89.62	0.9508
RepVit(3D version)	86.88	0.9010
Ours(3D version)	90.44	0.9430
Ours(2D)	**94.54**	**0.9685**

Table 7
The performance of using vessel segmentation in different strategies.

Strategy	ACC (%)	AUC
CVAM (Ours)	94.56	0.9685
VessSeg replaces CVAM	92.62	0.9627
Hybrid in the feature level	93.44	0.9664
Hybrid in the decision level	**94.81**	**0.9735**

Table 8
The experimental results of proposed VANet with and without extracting MCA regions in preprocessing.

Preprocessing setting	ACC (%)	AUC
extracting MCA	**94.54**	**0.9685**
w/o extracting MCA	86.07	0.9071

to evaluate the features learned by our proposed VANet. The clinical evaluation experiments include 1) detecting LVO on the input CTA images, the obtained feature heatmaps, and the combination of both; 2) evaluating whether the vessel feature maps obtained by VANet focus on the actual vessel correctly; 3) evaluating whether the asymmetry feature maps obtained by VANet are consistent with the actual asymmetry. The evaluation results are shown in Table 5. When only using the heatmaps generated by our model for LVO identification, the performance is close to the three radiologists' results only using CTA images. The combination of the heatmaps and CTA images can slightly improve the ACC of radiologists' diagnosis, which indicates the effectiveness of VANet's learned features. Additionally, the accuracy of vessel attention and asymmetry consistency evaluated by 3 radiologists exceed 95%, proving that our proposed VANet can accurately focus on vessels and asymmetry. These above results further illustrate the interpretability of our approach and validate its potential for assisting radiologists in clinical LVO diagnosis.

5.2. Advantages of VANet over 3D classification methods

In data preprocessing, we reconstruct the 3D CTA into a 2D image through projection and input it into the model, which may cause the loss of information. To explore whether the proposed method can also be better than some 3D analyses that can retain more spatial information, we also compare several 3D methods including a 3D version of DeepSymNet, a 3D version of RepVit, and a 3D version of VANet (Ours). These compared methods are also trained and tested on the same data split and the CTA scans are also registered to MNI space before feeding into models for fair comparison. As shown in Table 6, the results indicate that the performance of our method using MIP is superior to all compared 3D methods. This result is consistent with some existing studies such as (Wei et al., 2022) where MIP has shown superior performance compared to the original CTA scans in specific diagnostic tasks. In 3D CTA data, the proportion of vascular tissues is smaller, which makes it more difficult for the model to learn effective LVO features, resulting in poor performance of 3D methods. The above results imply that although using MIP may suffer from loss of spatial information, it enhances the performance of LVO identification.

5.3. The impact of combining vessel segmentation with CVAM

To validate whether combining vessel segmentation in our proposed VANet is better for LVO identification, we utilize SAM-VMNet (Zeng et al., 2024), a method for CTA vessel segmentation tasks based on foundation model, and perform several experiments that use some different strategies as follows. First, we replace the CVAM with the vessel segmentation results to guide the model's attention (called "VessSeg replaces CVAM"). Second, we fuse the vessel segmentation results and the output of CVAM using a concatenation operation to guide the model's attention (called "Hybrid in the feature level"). Third, we fuse the final outputs of the CVAM-guided model and the segmentation-guided model in the decision level using a trainable weighting module (called "Hybrid in the decision level"). The results are shown in Table 7. When using only segmentation results to replace the output of CVAM or fusing in the feature level, the performance is inferior to using CVAM only. This is due to that existing methods cannot effectively segment the vascular tissue, and poor segmentation results may bring noise, potentially negatively impacting the performance of the model. Fusing in the decision level slightly improves the performance, with ACC reaching 94.81% and AUC reaching 0.9735. We analyze the differences in classification samples between the model based on CVAM and the hybrid model based on fusing in the decision level and find that among the 20 misclassified samples by our proposed method with only CVAM, the hybrid model correctly classify 7 samples, indicating the usefulness of vessel segmentation for some hard samples. Therefore, in the future, we will explore a well-designed hybrid strategy to use vessel segmentation for better LVO identification.

5.4. The impact of extracting MCA regions

In the data preprocessing, we extract the MCA regions from the whole brain CTA. In order to explore the impact of extracting MCA regions, we conduct experiments to test the performance of VANet without extracting MCA regions in preprocessing. As shown in Table 8, the results indicates that the performance of extracting MCA regions is much higher than using original 3D CTA for MIP. After extracting the MCA regions, we believe that the model can more effectively focus on the vessels of interest, thereby improving the performance.

5.5. The impact of different projection algorithms

To explore and discuss the impact of different projection (also dimension reduction) strategies, we compare several commonly used projection algorithms in the field of medical image analysis, including MIP, Average Intensity Projection (AIP), and Minimum Intensity Projection (MinIP). As shown in Table 9, the ACC and AUC of MIP are both higher than those of AIP and MinIP. Besides, vessels appear with relatively higher intensity than other tissues in CTA, making MIP theoretically more suitable for highlighting vascular structures than AIP and MinIP. Therefore, MIP is selected as the projection method for the preprocessing in our method.

Table 9
The comparison of different projection algorithms.

Projection method	ACC (%)	AUC
MIP	94.54	0.9685
AIP	79.23	0.8248
MinIP	60.11	0.7039

Table 10
The comparison of different backbones used in our proposed method.

Backbone	Params(M)	ACC (%)	AUC
RepVit (Wang et al., 2024)(Ours)	**4.96**	**94.54**	**0.9685**
MoblieNetV3 (Howard et al., 2019)	6.07	93.44	0.9367
MedVit (Manzari et al., 2023)	32.88	94.54	0.9584
UniRepLKNet (Ding et al., 2024)	56.55	94.26	0.9625

Table 11
The comparison of different positions of VGFE.

VGFE position	ACC (%)	AUC
VGFE after Stage	**94.54**	**0.9685**
VGFE before Stage	94.26	0.9611

5.6. The impact of different backbones

We aim to develop an LVO identification method that achieves a well-balanced trade-off between performance and model size. Although models with fewer parameters often face limitations in performance, we have enhanced the model's capabilities through integrating clinical prior knowledge from multiple perspectives. We select several methods with varying parameter scales based on their performance from a diverse set of baselines (Howard et al., 2019; Manzari et al., 2023; Ding et al., 2024). The performance of different backbones is shown in Table 10. We can find that if we apply our modules to different backbones will result in varying degrees of performance improvement, proving the effectiveness of our designed modules. Compared to our proposed method, UniRepLKNet (Ding et al., 2024) achieves the same ACC as VANet while AUC is 0.0101 lower and the params of 56.55M which is larger than our VANet. Although the other two backbones (Manzari et al., 2023; Mao et al., 2022) achieve comparable performance, they still have parameters much higher than VANet. Therefore, selecting RepVit (Wang et al., 2024) as the backbone of VANet achieves a well-balanced trade-off between performance and model size.

5.7. The impact of VGFE's positions

We conduct experiments on the structure of VGFE and each stage of the model. We separately put VGFE before or after each stage to compare the performance of different structure. The experimental result is shown in Table 11. It is observed that placing VGFE after each stage achieves better performance. Hence, this structure has been adopted in the design of our VANet.

5.8. The impact of number of VGFE modules used

In order to prove the effectiveness of VGFE at different levels of the model, we start from the first stage and gradually add VGFE modules after each stage until all four stages are equipped with VGFE modules, to test how the model's performance changes. As shown in Table 12, the experimental results demonstrate that models with VGFE after each stage achieve the best performance. And as the number of VGFE decreases, the performance gradually decreases, which proves the effectiveness of VGFE in shallow and deep features. The results imply that proposed VGFE can help model learn effective vessel features at both shallow and deep level.

Table 12
The comparison of different number of VGFE used.

Stages with VGFE	ACC (%)	AUC
Stage 1	93.44	0.9569
Stage 1, 2	93.98	0.9514
Stage 1, 2, 3	94.26	0.9595
Stage 1, 2, 3, 4	**94.54**	**0.9685**

Table 13
The comparison of different AD calculating methods for deep asymmetry supervision, the ED represents Euclidean Distance, CS denotes Cosine Similarity and L1 denotes L1 Distance.

AD method	ACC (%)	AUC
Ours (MSE)	**94.54**	**0.9685**
ED	94.26	0.9593
CS	93.99	0.9587
L1	94.26	0.9624

Table 14
The performance of different weights of proposed deep asymmetry supervision and classification loss.

Weights setting (α : β)	ACC (%)	AUC
0.1 : 0.9	93.98	0.9597
0.3 : 0.7	94.26	0.9640
0.7 : 0.3	93.44	0.9595
0.9 : 0.1	93.17	0.9460
0.5 : 0.5	**94.54**	**0.9685**

5.9. The impact of AD calculation methods

How to calculate the similarity or distance between features and features flipped to quantitatively describe the asymmetry of features is significant. To explore the impact of different asymmetry calculation methods, we use different ways including Euclidean Distance, L1 Distance, Mean Square Error (MSE) and Cosine Similarity to quantitatively calculate asymmetry degree for our designed deep asymmetry supervision. Among them, the trend of cosine similarity is opposite to the others, that is, the higher the self-flip similarity, the stronger the symmetry, so we make some adjustments to change the trend through composite function in the implementation of the deep asymmetry supervision based on Cosine Similarity. The comparison of different AD methods is shown in Table 13. Results show that the MSE based AD calculating method achieves the best performance. Therefore, we utilize the MSE based asymmetry calculation method in our deep asymmetry supervision.

5.10. The impact of loss function weights

In order to explore the impact of different weights of loss functions on the performance of the model in the context of mixed loss functions, we conduct experiments by varying the weights assigned to the deep asymmetry supervision and the binary cross-entropy loss function, ensuring that their total weight sums to 1. The experimental results are shown in Table 14. The results indicate that when the weight of deep asymmetry supervision is too large, the model's performance decreases, even lower than when deep asymmetry supervision is not used. We attribute this to the smaller weight assigned to the main task, which is the classification task. When the weight of deep asymmetry supervision is smaller, there is a slight improvement in the model's performance. The best performance is achieved when the weights of deep asymmetry supervision and the classification loss are equal. Therefore, we set the weights of the two loss functions to 0.5 each.

5.11. Limitations and future work

There are several limitations of our proposed method. First, the dataset used in our methods is relatively small and from a limited number of medical institutions. Therefore, we will collect more data from different institutions and design transfer learning and fine-tuning methods based on existing foundation models to enhance the generalization ability of our approach. Second, there is information loss from original 3D CTA to preprocessed 2D image while the lost information may be helpful for clinical diagnosis. Therefore, we will explore how to develop a 3D LVO identification method with a small number of parameters and low computational complexity. Third, in some cases, small vessel occlusion may lead to wrong LVO identification of the proposed method. We will focus on both large and small vessels and develop a method to correctly distinguish small vessel occlusion from LVO.

6. Conclusion

In this paper, we propose a method named VANet for LVO identification without vessel segmentation from CTA, which is of significant importance for the treatment and prognosis of patients. Firstly, considering the characteristics of LVO and CTA data, we acquire coarse-grained vessel feature to direct the models attention towards vessel area. And then we design the Asymmetry Learning Module to extract effective asymmetry features. The experimental results on internal and external datasets show that our proposed method outperforms 11 state-of-the-art methods and achieves a well trade-off between performance and parameters. At the same time, the visualization results show that our method has good interpretability. Our proposed method can achieve efficient LVO identification for the external dataset, proving its good clinical application potential.

CRediT authorship contribution statement

Hulin Kuang: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Xinyuan Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin Liu:** Writing – review & editing, Validation, Methodology, Funding acquisition, Data curation, Conceptualization. **Shulin Liu:** Writing – review & editing, Validation, Resources, Investigation, Data curation, Conceptualization. **Shuai Yang:** Writing – review & editing, Validation, Resources, Investigation, Data curation, Conceptualization. **Weihua Liao:** Writing – review & editing, Validation, Resources, Investigation, Data curation, Conceptualization. **Wu Qiu:** Writing – review & editing, Validation, Resources, Methodology, Data curation, Conceptualization. **Guanghua Luo:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Data curation, Conceptualization. **Jianxin Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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